OpenClaw Peripheral Hardware Configuration

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- 1. Course Overview
- 2. Hardware Default Configuration File Description
  - 2.1 Configuration File Location
  - 2.2 Current Configuration File Coverage
- 3. DHT Current Configuration Features
  - 3.1 Verification Method
- 4. RGB Default Configuration
  - 4.1 Verification Method
- 5. OpenClaw Firmware Serial Port Default Configuration (Servo Verification)
  - 5.1 Current Default Values
  - 5.2 Modification Example
  - 5.3 Verification Method
- 6. OLED Default Configuration
  - 6.1 Current Default Values
  - 6.2 Scenarios Suitable for Modification
  - 6.3 Common Modifications
    - Changing Resolution
    - Changing I2C Address
    - Changing Rotation Direction
  - 6.4 Verification Method
- 7. Camera Hardware Configuration
  - 7.1 Configuration
  - 7.2 Checking Ports
  - 7.3 Photo Verification Method

1. Course Overview

This course introduces the default hardware-related configuration files used before operating OpenClaw, with a focus on how to modify the underlying default parameters based on the actual peripherals you have, and complete basic verification.

- Use underlying commands to verify whether modifications have taken effect

2. Hardware Default Configuration File Description

2.1 Configuration File Location

OpenClaw's current hardware default configuration file is located at:

```
/home/jetson/hardware_hal_demo/hardware_config.py
```

If you are only following the accompanying tutorial and using the bundled factory hardware, you usually do not need to modify it.

2.2 Current Configuration File Coverage

`hardware_config.py` mainly contains three types of default configurations:

- OpenClaw firmware serial port default parameters
- OLED default parameters
- CSI camera or USB camera

These default values are read by underlying command-line tools, such as:

- `rgb`
- `oled`
- `openclaw_adapter`
- `CAMERA_DEFAULTS`

Please note:

- **DHT and RGB are currently driven directly by the OpenClaw firmware**
- **OLED is driven by the motherboard's I2C pins**
- **CSI camera/USB camera requires checking the hardware connection**

3. DHT Current Configuration Features

3.1 Verification Method

```
dht   #Get sensor data
dht_oled  #Display sensor data on OLED
```

Normal terminal output should show the OLED displaying temperature and humidity data.



4. RGB Default Configuration

4.1 Verification Method

After saving, run the following in the terminal:

```
rgb color --color red #All lights turn red
rgb effect --effect breathe --color blue --speed 5 #Blue breathing light
```

Normal terminal output should show the RGB lights displaying correctly.

```
jetson@yahboom:~$ rgb color --color red
{"ok": true, "code": "ok", "message": "RGB 颜色已设置", "details": {"index": 255, "r": 255, "g": 0, "b": 0}}
jetson@yahboom:~$ rgb effect --effect breathe --color blue --speed 5
{"ok": true, "code": "ok", "message": "RGB 特效已设置: breathe", "details": {"effect": "breathe", "speed": 5, "color
": "blue"}}
jetson@yahboom:~$
```

5. OpenClaw Firmware Serial Port Default Configuration (Servo Verification)

5.1 Current Default Values

The current OpenClaw firmware serial port default parameters are:

- `port="/dev/ttyTHS1"`
- `baudrate=115200`

5.2 Modification Example

Edit the configuration file:

```
/home/jetson/hardware_hal_demo/hardware_config.py
```

For Jetson Orin NX, change the values as follows:

```
"port": "/dev/ttyTHS1",
"baudrate": 115200,
```

5.3 Verification Method

After saving, we recommend first testing with a servo or RGB using any underlying command, for example:

```
rgb color --color green
```

Or:

```
servo call-tool Servo1 --action center  #Servo 1 centers to 90 degrees
# All servos return to center position
servo call-tool ServoAll --action set_all --angle1 90 --angle2 90 --angle3 90 --angle4 90
```

If the commands execute successfully and control the peripherals properly, the serial port link is working.

```
jetson@yahboom:~$ servo call-tool Servo1 --action center
{"ok": true, "code": "ok", "message": "done", "device_id": "servo_1", "action": "center", "request_id": "", "state":
 {"angle": 90.0, "attached": true, "last_angle": 90.0, "min_angle": 0.0, "max_angle": 180.0}, "telemetry": {"mode":
"direct", "driver": "openclaw_servo", "servo_id": 1, "min_angle": 0.0, "max_angle": 180.0, "action": "set_angle", "p
arams": {}}, "retryable": false}
jetson@yahboom:~$ servo call-tool ServoAll --action set_all --angle1 90 --angle2 90 --angle3 90 --angle4 90
{"ok": true, "code": "ok", "message": "done", "device_id": "all", "action": "set_all", "request_id": "", "state": {"
angles": [90, 90, 90, 90]}, "telemetry": {"mode": "direct", "params": {"angle1": 90.0, "angle2": 90.0, "angle3": 90.
0, "angle4": 90.0}}, "retryable": false}
jetson@yahboom:~$
```

6. OLED Default Configuration

6.1 Current Default Values

The current OLED default parameters mainly include:

- `width=128`
- `height=32`
- `interface="i2c"`
- `i2c_bus="auto"`
- `i2c_bus_candidates=[1, 0, 7, 8]`
- `i2c_address=0x3C`
- `rotate=0`
- `invert=False`
- `contrast=255`

6.2 Scenarios Suitable for Modification

It is recommended to modify the OLED defaults in the following situations:

- You are using an OLED with a different resolution
- The I2C address is not `0x3C`
- The screen orientation needs to be flipped

6.3 Common Modifications

Changing Resolution

For example, change:

```
"width": 128,
"height": 32,
```

To:

```
"width": 128,
"height": 64,
```

Changing I2C Address

For example, change:

```
"i2c_address": 0x3C,
```

To:

```
"i2c_address": 0x3D,
```

Changing Rotation Direction

For example, change:

```
"rotate": 0,
```

To:

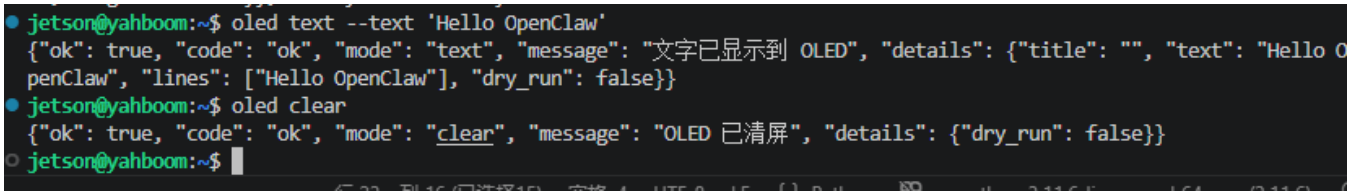
```
"rotate": 180,
```

6.4 Verification Method

After saving, run the following in the terminal:

```
oled text --text 'Hello OpenClaw'
oled clear
```

If the OLED can display text and clear the screen normally, the configuration has taken effect.



7. Camera Hardware Configuration

7.1 Configuration

Prerequisite: ensure that a CSI camera is connected. The current CSI camera link works mainly through the following two entry points:

If using a USB camera, you can modify the configuration in this file:

```
/home/jetson/hardware_hal_demo/hardware_config.py
```

```
# Camera default configuration
# backend: "csi" uses Jetson CSI camera; "usb" uses USB camera
CAMERA_DEFAULTS = {
    "backend": "usb",
    "usb_device": "/dev/video0",
    "usb_format": "auto",
    "width": 1280,
    "height": 720,
    "framerate": 30,
    "flip_method": 2,
    "sensor_id": 0,
}
```

Camera CLI command path:

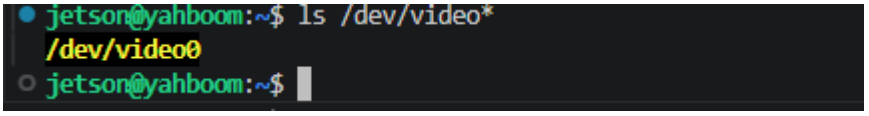
```
/home/jetson/hardware_hal_demo/cli/seewhat
```

7.2 Checking Ports

Enter in the terminal:

```
ls /dev/video*
```

Normally, a CSI camera should show one device, video0, and a USB camera should show video0 and video1 devices.



7.3 Photo Verification Method

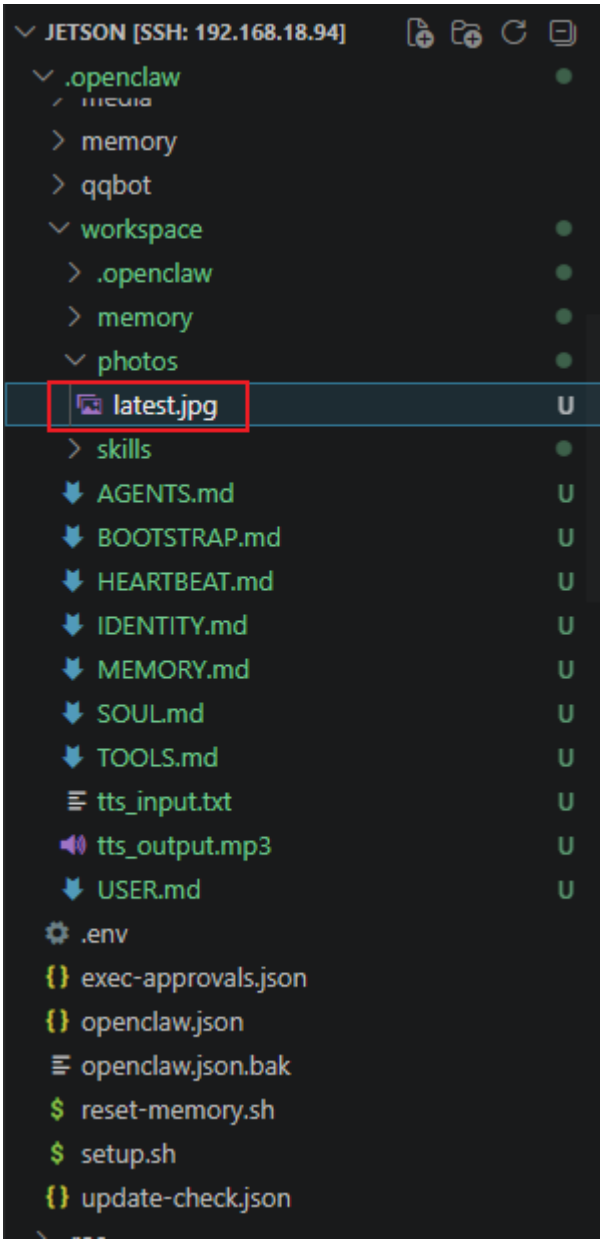
Enter the following command in the terminal to test the photo capture:

```
seewhat call-tool SeeWhat
```



If the photo is captured successfully, you should see the output file in the following directory:

```
/home/jetson/.openclaw/workspace/photos/latest.jpg
```


JETSON [SSH: 192.168.18.94]
.openclaw
memory
qqbot
workspace
.openclaw
memory
photos
latest.jpg
U
skills
AGENTS.md
BOOTSTRAP.md
HEARTBEAT.md
IDENTITY.md
MEMORY.md
SOUL.md
TOOLS.md
tts_input.txt
tts_output.mp3
USER.md
.env
exec-approvals.json
openclaw.json
openclaw.json.bak
reset-memory.sh
setup.sh
update-check.json